THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE CHAPTER TEST SET B

Chapter 6: **Pressure, Winds, Storms and Cyclones** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A force of 500 N acts perpendicularly on an area of 5 m². The pressure is: 1

(a) 200 N/m² **(b)** 100 N/m² **(c)** 50 N/m² **(d)** 2500 N/m²

2. The SI unit of pressure is: 1

(a) kilogram **(b)** newton **(c)** joule **(d)** pascal (N/m²)

3. The pressure of a liquid at the bottom increases with: 1

(a) shape of cap **(b)** height of the liquid column
(c) colour of liquid **(d)** weight of container

4. Pressure exerted by air is called: 1

(a) atmospheric pressure **(b)** static pressure only
(c) water pressure **(d)** liquid pressure

5. Wind blows from: 1

(a) north to south always **(b)** high pressure to low pressure
(c) hot to cold only **(d)** low pressure to high pressure

6. A land breeze blows: 1

(a) never **(b)** from land to sea at noon
(c) from land to sea at night **(d)** from sea to land at night

7. A storm is formed when warm moist air: 1

(a) sinks **(b)** freezes immediately
(c) stays still **(d)** rises, creating low pressure

8. Lower parts of thunderclouds carry mostly: 1

(a) magnetic poles **(b)** negative charges **(c)** positive charges **(d)** no charges

9. The eye of a cyclone is the region of: 1

(a) strongest wind **(b)** heaviest rain
(c) highest pressure **(d)** lowest pressure and calm wind

10. Local thunderstorm Kalboishakhi occurs in: 1

(a) Rajasthan only **(b)** West Bengal and nearby regions
(c) Kerala only **(d)** Ladakh

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. Why are overhead tanks placed at a height? 2

12. What is atmospheric pressure? 2

13. Why are we not crushed by atmospheric pressure? 2

14. Explain how a rubber sucker sticks to a wall. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Explain sea breeze and land breeze with the direction of wind in each. 3

16. Explain why it is safer to keep doors and windows open during a storm. 3

17. Describe the steps of formation of a storm. 3

OR

How are charges produced in clouds during a thunderstorm? 3

18. Write three safety measures during a thunderstorm. 3

OR

Describe a lightning conductor and how it protects a building. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. Why do high-speed winds reduce air pressure? Explain with the balloon activity and the roof example. 5

OR

Explain the formation of a storm and a thunderstorm step by step. 5

20. Describe the process of lightning formation. State four safety rules. 5

OR

Explain how a lightning conductor works with a diagram. 5

END OF QUESTION PAPER — Total: 40 Marks

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